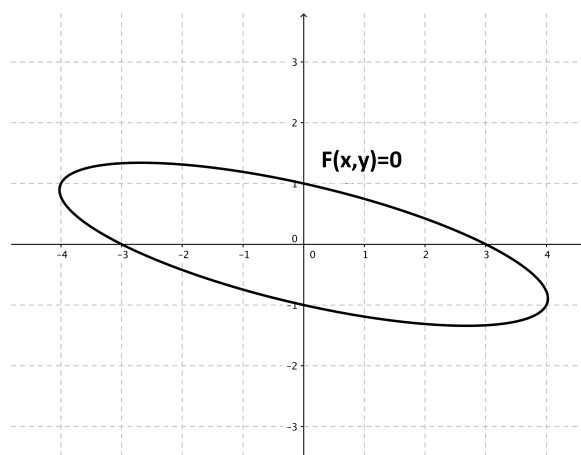


Problem 1

Problem 1

Problem 1 On the graph below, sketch the tangent lines at $x = 0$. Then, explain why both the x -coordinate and the y -coordinate are generally needed to find the slope of the tangent line at a point on the graph of an equation of the form $F(x, y) = 0$



Explanation. Given the equation $F(x, y) = 0$, its graph may not pass the “vertical line test”. As illustrated in the figure below, fixing a value for x (in this case $x = 0$) will not typically specify a unique point on the graph because there may be more than one corresponding y -value. That is why you need to specify both the x and the y coordinate, to make sure that you are giving the coordinates for a unique point on the graph. In this case, there are two tangent lines to the curve at $x = 0$. One passes through the point $(0, 1)$ and the other through $(0, -1)$.

Problem 1

